

Security Audit

Igna (Token)

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CAUTION

THIS DOCUMENT IS A SECURITY AUDIT REPORT AND MAY CONTAIN CONFIDENTIAL INFORMATION. THIS INCLUDES IDENTIFIED VULNERABILITIES AND MALICIOUS CODE WHICH COULD BE USED TO COMPROMISE THE PROJECT. THIS DOCUMENT SHOULD ONLY BE FOR INTERNAL USE UNTIL ISSUES ARE RESOLVED. ONCE VULNERABILITIES ARE REMEDIATED, THIS REPORT CAN BE MADE PUBLIC. THE CONTENT OF THIS REPORT IS OWNED BY HASHLOCK PTY LTD FOR USE OF THE CLIENT.

Executive Summary

The Igna team partnered with Hashlock to conduct a security audit of their token. Hashlock manually and proactively reviewed the code in order to ensure the project's team and community that the deployed contracts are secure.

Project Context

Igna Crypto is a comprehensive digital asset platform that offers innovative trading and investment solutions for the cryptocurrency market. The platform emphasizes robust security, advanced trading tools, and educational resources, enabling both new and seasoned investors to navigate the fast-paced world of digital assets with confidence. By leveraging cutting-edge technology and market insights, Igna Crypto aims to simplify portfolio management and empower users to capitalize on emerging opportunities in the crypto space.

Project Name: Igna

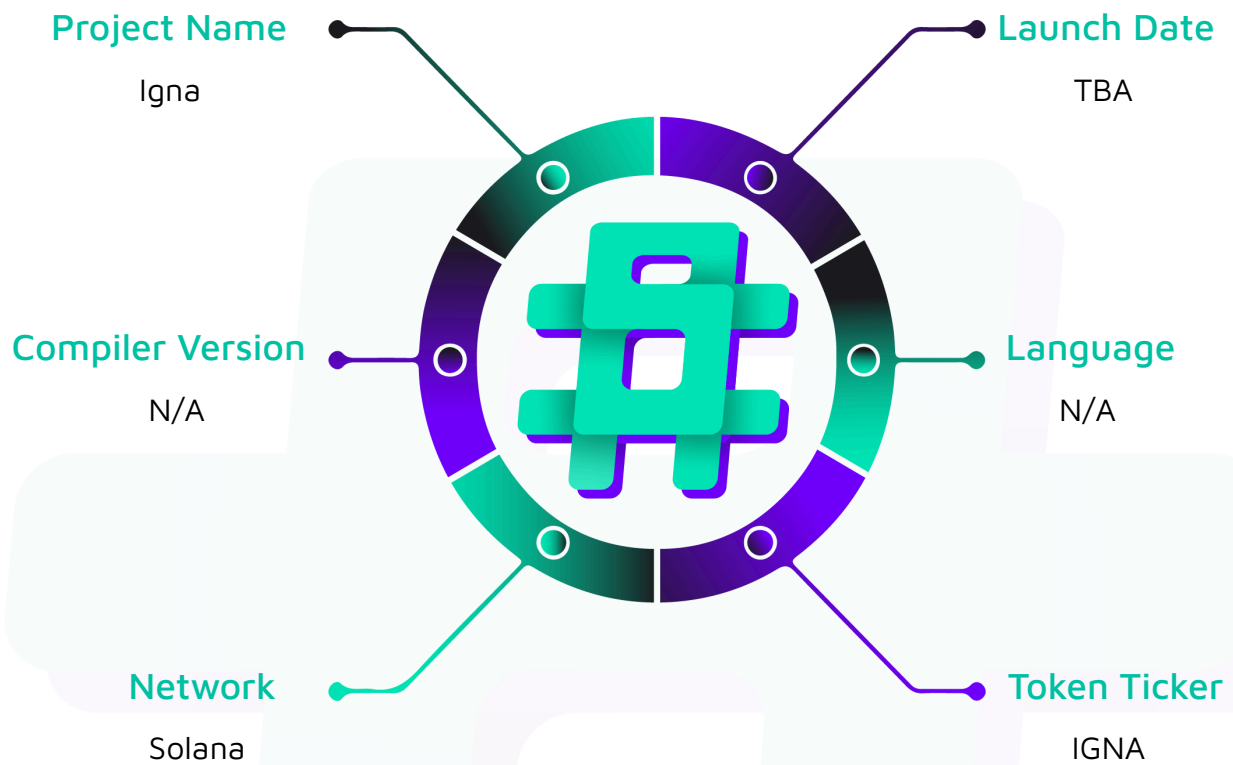
Project Type: Token

Compiler Version: N/A

Website: <https://www.igna-crypto.com/>

Logo:



Visualised Context:

Project Visuals:



The image displays the IGNA Token project website. The header features the IGNA logo and a navigation menu with links: HOME, FEATURE, ICO CHART, ROADMAP, NEWS, and CONTACT. The main banner shows a collage of images: a large green industrial machine processing waste, a stack of gold coins, and a small green plant growing from a stack of coins. The text 'IGNA TOKEN' is overlaid on the collage. Below the banner, the headline reads 'TURN POLLUTION INTO SOLUTION: TRANSFORM WASTE INTO WEALTH'. A sub-headline states: 'Join the revolutionary movement to turn plastic waste into a profitable investment with the help of our pioneering cryptocurrency.' To the left of this text are three IGNA tokens. A blue box contains the following text: 'By investing in IGNA, you're not just buying crypto — you are contributing to the creation of pyrolysis plants that turn waste plastic into oil, reducing global pollution. Your investment supports worldwide adoption of this technology, promoting a cleaner environment. Join us by buying our crypto and make a positive impact today!'. Below this, the text 'Participate in the project' is displayed. A white box shows the price: '1 IGNA = \$0.01991 USDC'. A small note below the price states: '* based on the current price of the [DEX-TRADE.com](#) exchange'. At the bottom, there are two buttons: 'Purchase IGNA >' and 'Read White Paper >'.

IGNA

HOME FEATURE ICO CHART ROADMAP NEWS CONTACT

IGNA TOKEN

TURN POLLUTION INTO SOLUTION:
TRANSFORM WASTE INTO WEALTH

Join the revolutionary movement to turn plastic waste into a profitable investment with the help of our pioneering cryptocurrency.

By investing in IGNA, you're not just buying crypto — you are contributing to the creation of pyrolysis plants that turn waste plastic into oil, reducing global pollution. Your investment supports worldwide adoption of this technology, promoting a cleaner environment. Join us by buying our crypto and make a positive impact today!

Participate in the project

1 IGNA = \$0.01991 USDC

* based on the current price of the [DEX-TRADE.com](#) exchange

Purchase IGNA >

Read White Paper >

Audit Scope

We at Hashlock audited the Igna token within the Igna project, the scope of work included a comprehensive review of the token.

Description	IGNA Token Review
Platform	Solana
Audit Date	March, 2025
Token Name	IGNA
Token Address	ignFri3b6VJxpov64XMa2d81dWrHrMAJLWkFAM6ER4f

Security Rating

After Hashlock's Audit, we found the token to be **"Secure"**. We initially identified a vulnerability that has since been addressed.



Not Secure

Vulnerable

Secure

Hashlocked

The 'Hashlocked' rating is reserved for projects that ensure ongoing security via bug bounty programs or on chain monitoring technology.

All issues uncovered during automated and manual analysis were meticulously reviewed and applicable vulnerabilities are presented in the [Audit Findings](#) section. The list of audited assets is presented in the [Audit Scope](#) section and the project's contract functionality is presented in the [Intended Smart Contract Functions](#) section.

The finding we have identified has now been resolved.

Hashlock found:

1 Low severity vulnerability

Caution: Hashlock's audits do not guarantee a project's success or ethics, and are not liable or responsible for security. Always conduct independent research about any project before interacting.

Intended Smart Contract Functions

Claimed Behaviour	Actual Behaviour
IGNA Token - Simple SPL token	Contract achieves this functionality.

Code Quality

This audit scope involves the token of the Igna project, as outlined in the Audit Scope section. All contracts, libraries, and interfaces mostly follow standard best practices and to help avoid unnecessary complexity that increases the likelihood of exploitation, however, some refactoring was recommended.

Audit Resources

We were given the Igna project token in the form of a Token address.

As mentioned above, code parts are well commented. The logic is straightforward, and therefore it is easy to quickly comprehend the programming flow as well as the complex code logic. The comments are helpful in providing an understanding of the protocol's overall architecture.

Dependencies

As per our observation, the libraries used in this smart contracts infrastructure are based on well-known industry standard open source projects.

Severity Definitions

The severity levels assigned to findings represent a comprehensive evaluation of both their potential impact and the likelihood of occurrence within the system. These categorizations are established based on Hashlock's professional standards and expertise, incorporating both industry best practices and our discretion as security auditors. This ensures a tailored assessment that reflects the specific context and risk profile of each finding.

Significance	Description
High	High-severity vulnerabilities can result in loss of funds, asset loss, access denial, and other critical issues that will result in the direct loss of funds and control by the owners and community.
Medium	Medium-level difficulties should be solved before deployment, but won't result in loss of funds.
Low	Low-level vulnerabilities are areas that lack best practices that may cause small complications in the future.
Gas	Gas Optimisations, issues, and inefficiencies.
QA	Quality Assurance (QA) findings are informational and don't impact functionality. Supports clients improve the clarity, maintainability, or overall structure of the code.

Status Definitions

Each identified security finding is assigned a status that reflects its current stage of remediation or acknowledgment. The status provides clarity on the handling of the issue and ensures transparency in the auditing process. The statuses are as follows:

Significance	Description
Resolved	The identified vulnerability has been fully mitigated either through the implementation of the recommended solution proposed by Hashlock or through an alternative client-provided solution that demonstrably addresses the issue.
Acknowledged	The client has formally recognized the vulnerability but has chosen not to address it due to the high cost or complexity of remediation. This status is acceptable for medium and low-severity findings after internal review and agreement. However, all high-severity findings must be resolved without exception.
Unresolved	The finding remains neither remediated nor formally acknowledged by the client, leaving the vulnerability unaddressed.

Audit Findings

Low

[L-01] Token - Centralization Risk Due to Token Concentration in Private Wallet

Description

After the deployment of the token, a large amount of tokens were transferred to different single wallets. This creates a significant centralization risk because the security of the entire system becomes dependent on the safety of this wallet's private key. If the private key of this wallet is compromised, an attacker could gain control over the majority of the tokens, potentially leading to unauthorized transactions, market manipulation, or a complete loss of trust in the system.

Recommendation

It's recommended to use multi-sig wallets to enhance security.

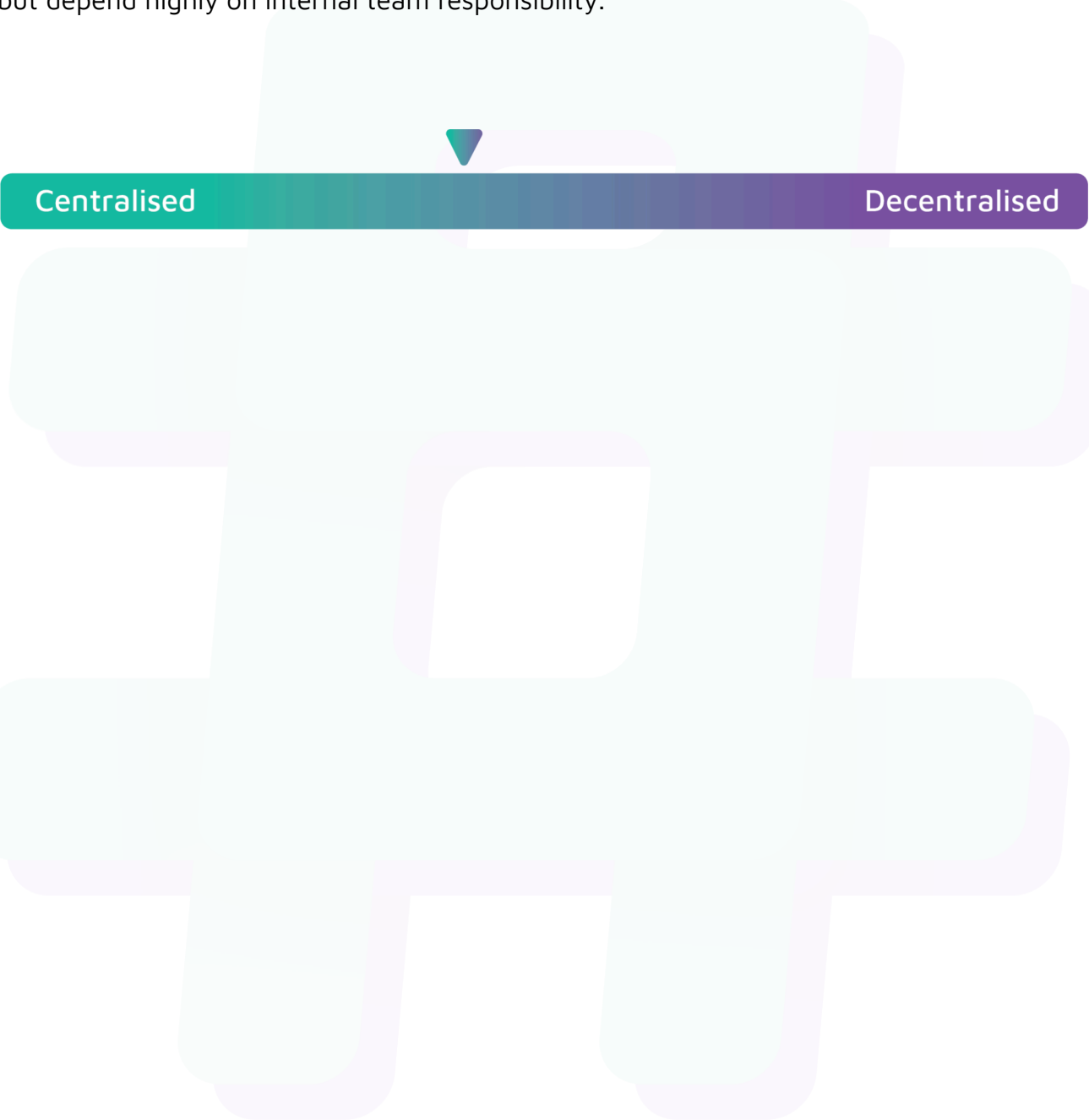
Status

Resolved

Centralisation

The Igna project values security and utility over decentralisation.

The owner executable functions within the protocol increase security and functionality but depend highly on internal team responsibility.



Centralised

Decentralised

Conclusion

After Hashlock's analysis, the Igna project seems to have a sound and well-tested code base, now that our finding has been resolved. Overall, most of the code is correctly ordered and follows industry best practices. The code is well commented on as well. To the best of our ability, Hashlock is not able to identify any further vulnerabilities.



Our Methodology

Hashlock strives to maintain a transparent working process and to make our audits a collaborative effort. The objective of our security audits is to improve the quality of systems and upcoming projects we review and to aim for sufficient remediation to help protect users and project leaders. Below is the methodology we use in our security audit process.

Manual Code Review:

In manually analysing all of the code, we seek to find any potential issues with code logic, error handling, protocol and header parsing, cryptographic errors, and random number generators. We also watch for areas where more defensive programming could reduce the risk of future mistakes and speed up future audits. Although our primary focus is on the in-scope code, we examine dependency code and behaviour when it is relevant to a particular line of investigation.

Vulnerability Analysis:

Our methodologies include manual code analysis, user interface interaction, and white box penetration testing. We consider the project's website, specifications, and whitepaper (if available) to attain a high-level understanding of what functionality the smart contract under review contains. We then communicate with the developers and founders to gain insight into their vision for the project. We install and deploy the relevant software, exploring the user interactions and roles. While we do this, we brainstorm threat models and attack surfaces. We read design documentation, review other audit results, search for similar projects, examine source code dependencies, skim open issue tickets, and generally investigate details other than the implementation.

Documenting Results:

We undergo a robust, transparent process for analysing potential security vulnerabilities and seeing them through to successful remediation. When a potential issue is discovered, we immediately create an issue entry for it in this document, even though we have not yet verified the feasibility and impact of the issue. This process is vast because we document our suspicions early even if they are later shown to not represent exploitable vulnerabilities. We generally follow a process of first documenting the suspicion with unresolved questions, and then confirming the issue through code analysis, live experimentation, or automated tests. Code analysis is the most tentative, and we strive to provide test code, log captures, or screenshots demonstrating our confirmation. After this, we analyse the feasibility of an attack in a live system.

Suggested Solutions:

We search for immediate mitigations that live deployments can take and finally, we suggest the requirements for remediation engineering for future releases. The mitigation and remediation recommendations should be scrutinised by the developers and deployment engineers, and successful mitigation and remediation is an ongoing collaborative process after we deliver our report, and before the contract details are made public.

Disclaimers

Hashlock's Disclaimer

Hashlock's team has analysed these smart contracts in accordance with the best industry practices at the date of this report, in relation to: cybersecurity vulnerabilities and issues in the smart contract source code, the details of which are disclosed in this report, (Source Code); the Source Code compilation, deployment, and functionality (performing the intended functions).

Due to the fact that the total number of test cases is unlimited, the audit makes no statements or warranties on the security of the code. It also cannot be considered as a sufficient assessment regarding the utility and safety of the code, bug-free status, or any other statements of the contract. While we have done our best in conducting the analysis and producing this report, it is important to note that you should not rely on this report only. We also suggest conducting a bug bounty program to confirm the high level of security of this smart contract.

Hashlock is not responsible for the safety of any funds and is not in any way liable for the security of the project.

Technical Disclaimer

Smart contracts are deployed and executed on a blockchain platform. The platform, its programming language, and other software related to the smart contract can have their own vulnerabilities that can lead to attacks. Thus, the audit can't guarantee the explicit security of the audited smart contracts.

About Hashlock

Hashlock is an Australian-based company aiming to help facilitate the successful widespread adoption of distributed ledger technology. Our key services all have a focus on security, as well as projects that focus on streamlined adoption in the business sector.

Hashlock is excited to continue to grow its partnerships with developers and other web3-oriented companies to collaborate on secure innovation, helping businesses and decentralised entities alike.

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#hashlock.